

Zekun Sun
zekun.sun@yale.com
<https://zekun-sun.github.io/>

Academic Appointment

2023– Postdoctoral Associate, [Action, Computation and Thinking Lab](#), Yale University (Advisor: [Samuel D. McDougle](#))

Education

2017–23 **Ph.D.**, Cognitive Psychology, Johns Hopkins University
(Advisor: [Chaz Firestone](#))

2013-16 **M.S.**, Cognitive Neuroscience, Chinese Academy of Sciences

2009-13 **LL.B.**, International Politics, Nanjing University

Publications

 [Google Scholar](#)

Journal Articles

- J1. **Sun, Z.** & McDougle, S. D. Perceiving event structure in brief actions. *Cognitive Psychology* (in second revision).
- J2. **Sun, Z.**, Firestone, C. & Hafri, A. The psychophysics of compositionality: Relational scene perception occurs in a canonical order. *Cognitive Psychology*. in press (2025).
- J3. **Sun, Z.** & Firestone, C. Caricaturing shapes in visual memory. *Psychological Science* **35**, 722–735 (2024).
- J4. **Sun, Z.** & Firestone, C. Beautiful on the inside: Aesthetic preferences and the skeletal complexity of shapes. *Perception* **51**, 904–918 (2022).
- J5. **Sun, Z.** & Firestone, C. Seeing and speaking: How verbal “description length” encodes visual complexity. *Journal of Experimental Psychology: General* **151**, 82 (2022).
- J6. **Sun, Z.** & Firestone, C. Curious objects: How visual complexity guides attention and engagement. *Cognitive science* **45**, e12933 (2021).
– **Top-cited article in *Cognitive Science*, 2022.**

- J7. **Sun, Z.** & Firestone, C. Optimism and pessimism in the predictive brain. *Trends in Cognitive Sciences* **24**, 683–685 (2020).
- J8. **Sun, Z.** & Firestone, C. The dark room problem. *Trends in cognitive sciences* **24**, 346–348 (2020).
- J9. Fan, L., Sun, Y.-B., **Sun, Z.**, Wang, N., Luo, F., Yu, F. & Wang, J.-Y. Modulation of auditory sensory memory by chronic clinical pain and acute experimental pain: a mismatch negativity study. *Scientific Reports* **8**, 15673 (2018).
- J10. **Sun, Z.**, Wang, J.-Y. & Luo, F. Experimental pain induces attentional bias that is modified by enhanced motivation: An eye tracking study. *European Journal of Pain* **20**, 1266–1277 (2016).

Working papers

- W1. **Sun, Z.**, Bai, D., Scholl, B. J. & McDougle, S. D. *Visuomotor adaptation via featureless objects.* ([full manuscript](#)).
- W2. **Sun, Z.** & McDougle, S. D. *Motor planning is obligatorily biased by task-irrelevant objects.*
- W3. **Sun, Z.**, Xie, Z. & McDougle, S. D. *Novel skills recruit motor abstractions.*

Presentations

Invited Talks

- I1. Perception and Action Seminar, Brown University. Sept. 2025.
Title: Representational abstraction in visual and motor cognition.
- I2. The Skilled Action and Memory Lab, University of Birmingham. July 2025.
Title: Novel skills recruit motor abstractions.
- I3. Cognitive Current Works, Yale University. Feb. 2025.
Title: Novel skills recruit motor abstractions.
- I4. The Action, Cognition and Thinking Lab, Yale University. June 2022.
Title: Cognitive consequences of visual complexity.
- I5. Behrmann Lab, Carnegie Mellon University. Apr. 2022.
Title: Cognitive consequences of visual complexity.
- I6. The Computational Vision and Learning Lab, University of California, Los Angeles. Apr. 2022.
Title: Cognitive consequences of visual complexity.
- I7. The Language and Cognition Lab, Stanford University. Jan. 2022.
Title: Cognitive consequences of visual complexity.

- I8. Cognitive Tools Lab, University of California, San Diego. Oct. 2021.
Title: Cognitive consequences of visual complexity.
- I9. The Computation and Language Lab, University of California, Berkeley.
Oct. 2021.
Title: Cognitive consequences of visual complexity.

Talks

- T1. **Sun, Z.** & D., M. S. *Dissociating external features from internal structures in visual segmentation of actions.* The 25nd Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2025.
- T2. Hafri, A., **Sun, Z.** & Firestone, C. *The psychophysics of compositionality: Relational scene perception occurs in a canonical order.* The 24nd Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2024.
- T3. **Sun, Z.**, Bi, W., Yildirim, I. & McDougale, S. D. *Breaking down a golf swing: Spatio-temporal dynamics of visual motion underlie high-level structuring of observed actions.* The 24nd Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2024.
- T4. **Sun, Z.**, Han, S. & Firestone, C. *The evolution of complexity in visual memory.* The 21th annual meeting of the Vision Sciences Society (online). Nov. 2021.
- T5. Yu, Q. & **Sun, Z.** *Looking unique.* the 29th annual meeting for Object Perception, Attention, and Memory (online). Nov. 2021.
- T6. **Sun, Z.** & Firestone, C. *Speaking about seeing: How verbal descriptions encode visual complexity.* The 27th annual meeting of Object Perception, Attention, and Memory (Montreal, Canada). Nov. 2019.
– **Best Talk Award.**
- T7. **Sun, Z.** & Firestone, C. *Curious objects: Preattentive processing of object complexity.* The 18th annual meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). Nov. 2018.

Posters

- P1. **Sun, Z.**, Bai, D., Scholl, B. J. & McDougale, S. D. *Motor Adaptation via Pure Visual Change.* Neural Control of Movement 2025 (Panama City, Panama). July 2025.
- P2. **Sun, Z.** & McDougale, S. D. *Novel motor skills recruit visuomotor abstractions.* The 25nd Annual Meeting of the Vision Sciences Society (St. Pete Beach, FL, USA). May 2025.

- P3. Kang, S., **Sun, Z.** & McDougale, S. D. *Investigating Structural Learning in a De Novo Motor Skill.* Society for Neuroscience 2024 (Chicago, IL, USA). Oct. 2024.
- P4. **Sun, Z.** & McDougale, S. D. *Goal uncertainty biases memory for observed actions.* Vision Sciences Society 2024 (St. Pete Beach, FL, USA). May 2024.
- P5. **Sun, Z.** & McDougale, S. D. *Repelling Effects of Distractors Persist in Imitated Movements.* Society for Neuroscience 2024 (Chicago, IL, USA). Oct. 2024.
- P6. Han, S., **Sun, Z.** & Firestone, C. *Caricaturing shape in visual memory.* Vision Sciences Society 2023 (St. Pete Beach, FL, USA). May 2023.
- P7. **Sun, Z.**, Hafri, A. & Firestone, C. *How to build a scene: Relational representations are constructed in a canonical order.* Vision Sciences Society 2023 (St. Pete Beach, FL, USA). May 2023.
- P8. Hafri, A., **Sun, Z.** & Firestone, C. *Sequential construction of visual relations.* Vision Sciences Society 2022 (St. Pete Beach, FL, USA). May 2022.
- P9. **Sun, Z.**, Yu, Q. & Firestone, C. *How to look unique.* Vision Sciences Society 2022 (St. Pete Beach, FL, USA). May 2022.
- P10. **Sun, Z.** & Firestone, C. *The evolution of complexity in visual memory.* Object Perception, Attention, and Memory 2021 (Online). Nov. 2021.
- P11. Halberda, J., Yu, Q., **Sun, Z.** & Firestone, C. *Not too simple, not too complex: The Goldilocks principle drives discrimination and search.* Vision Sciences Society 2020 (Online). June 2020.
- P12. **Sun, Z.** & Firestone, C. *The simple and the beautiful: An aesthetic preference for medially complex stimuli.* Vision Sciences Society 2020 (Online). June 2020.
- P13. **Sun, Z.** & Firestone, C. *Speaking about seeing: How verbal descriptions encode visual complexity.* Vision Sciences Society 2019 (St. Pete Beach, FL, USA). May 2019.
– **Student Travel Award.**

Awards & Honors

2022	G. Stanley Hall's Scholar Award Johns Hopkins University
2021	Clark Collaborative Award Johns Hopkins University
2021	The Mary Ainsworth Award Johns Hopkins University

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| 2019 | The Robert S. Waldrop Junior Investigator's Award
Johns Hopkins University |
| 2019 | OPAM Best Talk Award
Object Perception, Attention, and Memory |
| 2019 | Elsevier/Vision Research Travel Award
Vision Sciences Society |
| 2018/2019 | Collaborative Research Award
Johns Hopkins University |
| 2015/2016 | Graduate Student Award
Chinese Academy of Sciences |

Teaching

Johns Hopkins University

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| 2020 | Teaching Assistant, Real World Human Data |
| 2019 | Teaching Assistant, Methods in Experimental Psychology |
| 2019 | Teaching Assistant, Introduction to Cognitive Psychology |
| 2018 | Teaching Assistant, Introduction to Social Psychology |

Academic Advising

Graduate

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| 2025 | Zhiran Xie (Master student in Computational Biology), Yale University. Research project on motor control and neural networks, research assistant |
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Undergraduate

- 2025 Emma Kohlmayer (Art History and Cognitive Science), Yale University. Research project on action and perception and senior thesis.
- 2024 Zack Haaland (Cognitive Science), Yale University. Senior thesis.
- 2022/2023 Subin Han, Johns Hopkins University. Psychology project internship that resulted in Hopkins undergraduate research award (PURA) and a co-authored publication in *Psychological Science*.

Academic Service

Intern Junior Editor

- 2025 – 2026 Journal of Experimental Psychology:
Human Perception and Performance

Journal Reviewer

Attention, Perception, and Psychophysics
iPerception
iScience
Journal of Cognitive Neuroscience
Journal of Experimental Psychology: Human Perception and
Performance
Journal of Experimental Psychology: General
Philosophical Transactions of the Royal Society B
Perception
Synthese
Visual Cognition

Other experience

- 2020 Early Career Colloquium Selection Committee
— Johns Hopkins University
Role: Student Organizer

- 2019 Summer Internship Program
 — Johns Hopkins University
 Role: Mentor, Vision Group
 Project: Shape bias and complexity bias
 Mentee: Subin Han, Cognitive Science Department
- 2019 Department Colloquium Committee
 — Johns Hopkins University
 Role: Student Organizer

References

Samuel D. McDougale

Department of Psychology
Yale University
samuel.mcdougale@yale.edu

Alon Hafri

Linguistics & Cognitive Science Department
University of Delaware
alon@udel.edu

Chaz Firestone

Psychological & Brain Sciences Department
Johns Hopkins University
chaz@jhu.edu

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